

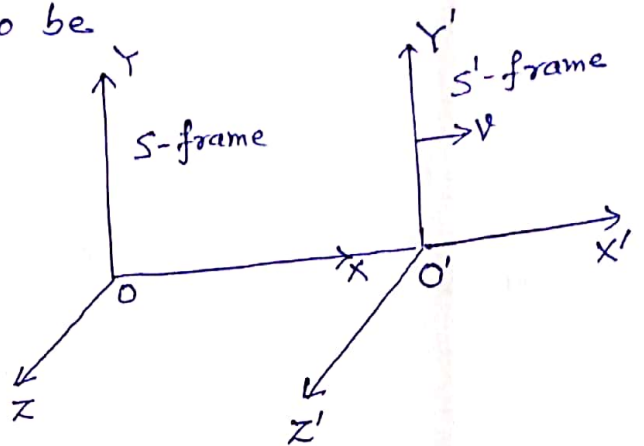
VELOCITY ADDITION

Let us consider something moving relative to both S and S' frame of references. An observer in S measures its three velocity components to be

$$V_x = \frac{dx}{dt}$$

$$V_y = \frac{dy}{dt}$$

$$V_z = \frac{dz}{dt}$$



while an observer in S' -frame measures

$$V'_x = \frac{dx'}{dt'}$$

$$V'_y = \frac{dy'}{dt'}$$

$$V'_z = \frac{dz'}{dt'}$$

We have from Lorentz transformation equation

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - \frac{v}{c^2}x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

By differentiating these Lorentz transformation we have

$$dx' = \frac{dx - vdt}{\sqrt{1 - v^2/c^2}}$$

$$dy' = dy$$

$$dz' = dz$$

$$dt' = \frac{dt - \frac{v}{c^2}dx}{\sqrt{1 - \frac{v^2}{c^2}}}$$

So,

$$\begin{aligned}V_x' &= \frac{dx'}{dt'} \\&= \frac{\frac{dx - v dt}{\sqrt{1 - v^2/c^2}}}{\frac{dt - \frac{v}{c^2} dx}{\sqrt{1 - v^2/c^2}}} \\&= \frac{dx - v dt}{dt - \frac{v}{c^2} dx} \\&= \frac{\frac{dx}{dt} - v}{1 - \frac{v}{c^2} \frac{dx}{dt}}\end{aligned}$$

$$\boxed{V_x' = \frac{V_x - v}{1 - \frac{v}{c^2} V_x}}$$

————— ①.

Now

$$\begin{aligned}V_y' &= \frac{dy'}{dt'} \\&= \frac{\frac{dy}{dt - \frac{v}{c^2} dx}}{\frac{\sqrt{1 - \frac{v^2}{c^2}}}{\sqrt{1 - \frac{v^2}{c^2}}}} \\&= \frac{dy \sqrt{1 - \frac{v^2}{c^2}}}{dt - \frac{v}{c^2} dx} \\&= \frac{\frac{dy}{dt} \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2} \frac{dx}{dt}}\end{aligned}$$

$$\boxed{V_y' = \frac{V_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2} V_x}}$$

————— ②

Similarly

$$\boxed{V_z' = \frac{V_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2} V_x}}$$

————— ③

Equations (1), (2) & (3) constitute the relativistic velocity transformations. When $v \ll c$, these equations reduce to Galilean transformations

$$V_x' = V_x - v$$

$$V_y' = V_y$$

$$V_z' = V_z$$

To express V_x, V_y, V_z in terms of V_x', V_y', V_z' , it is needed only to replace v by $-v$ and exchange primed and unprimed quantities in eqs. (1) to (3). Hence

$$V_x = \frac{V_x' + v}{1 + \frac{v}{c^2} V_x'} \quad \text{--- (4)}$$

$$V_y = \frac{V_y' \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c^2} V_x'} \quad \text{--- (5)}$$

$$V_z = \frac{V_z' \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c^2} V_x'} \quad \text{--- (6)}$$

Discussion

- (i) If $V_x' = c$ i.e. if light is emitted in the moving frame S' in the direction of motion relative to S , the observer in S will measure the velocity

$$\begin{aligned} V_x &= \frac{V_x' + v}{1 + \frac{v}{c^2} V_x'} \\ &= \frac{c + v}{1 + \frac{v}{c^2} c} = \frac{c + v}{1 + \frac{v}{c}} \\ &= c \end{aligned}$$

Thus, if an object moves with velocity c with respect to other, then their relative velocity is always c whatever may be the velocity of the other.

- (ii) When $V_x' = v = c$, then
- $$V_x = \frac{c + c}{1 + \frac{c^2}{c^2}} = c$$

This shows that the addition of velocity of light to the velocity of light merely reproduces the velocity of light.

It concludes that the velocity of light is an absolute constant.

(iii) If a light pulse propagates along y' -axis in S' -frame with a velocity c ($c = U_y$), then the components of velocity of light in S -frame are

$$U_y = \frac{U_y \sqrt{1 - v^2/c^2}}{1 + \frac{v}{c^2} U_{x'}} \\ = c \sqrt{1 - v^2/c^2} \quad (\text{because } U_{x'} = 0)$$

$$\& \quad U_x = v$$

$$U_z = 0$$

Consequently, a ray of light which is normal to the x' -axis in S' -frame has a different direction in S -frame.

But, the magnitude of the speed of light remains c .

$$\text{Here } V = \sqrt{U_x^2 + U_y^2} \\ = \sqrt{v^2 + c^2(1 - \frac{v^2}{c^2})} \\ = \sqrt{v^2 + c^2 - v^2} = c.$$

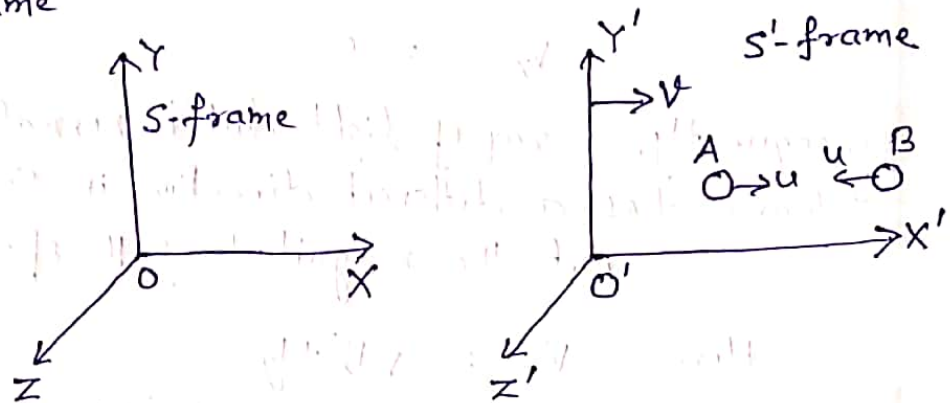
This change of the direction of the rays of light accounts for the aberration of starlight which is an observed seasonal change in the position of stars. It appears due to the Earth's orbital motion.

(iv) The components of the velocity along the y - and z -axes also change, because the time intervals change and $dt \neq dt'$.

(v) If $v = c$, the radicands become equal to zero and in this case also, the formulae do not have physical meaning. It means that there is no reference frame in which the a photon is at rest.

VARIATION OF MASS WITH VELOCITY

Let us consider two frame of references S & S' . S' is moving with a uniform velocity v relative to S in the positive x -direction. Let us suppose in S' -frame, two exactly similar elastic balls A and B each of mass m approach each other at equal speeds (i.e. u and $-u$). They collide with each other and coalesce into one body. Considering conservation of momentum in S' -frame



$$\text{Initial momentum} = mu - mu = 0$$

$$\text{Final momentum} = 0$$

Thus, the coalesced mass will be at rest in frame S' . Let us consider the collision with respect to S -frame. Let u_1 and u_2 be the velocities of the ball A and B observed by the observer in S -frame. So,

$$u_1 = \frac{u+v}{1 + \frac{v}{c^2}u} \quad \text{--- (1)}$$

$$u_2 = \frac{-u+v}{1 - \frac{v}{c^2}u} \quad \text{--- (2)}$$

After collision velocity of the coalesced mass is v relative to S -frame. Let the mass of the ball A moving with velocity u_1 be m_1 , and that of ball B moving with velocity u_2 be m_2 in the frame S .

Considering conservation of momentum in S -frame, we have

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

Substituting for u_1 and u_2 from equation (1) and (2), we have

$$m_1 \left(\frac{u+v}{1 + \frac{uv}{c^2}} \right) + m_2 \left(\frac{-u+v}{1 - \frac{uv}{c^2}} \right) = (m_1 + m_2) v$$

$$\text{or } m_1 \left[\frac{u+v}{1 + \frac{uv}{c^2}} - v \right] = m_2 \left[v - \frac{-u+v}{1 - \frac{uv}{c^2}} \right]$$

$$\text{or } m_1 \left[\frac{u+v - v + \frac{uv^2}{c^2}}{1 + \frac{uv}{c^2}} \right] = m_2 \left[\frac{v - \frac{-u+v}{1 - \frac{uv}{c^2}}}{1 - \frac{uv}{c^2}} \right]$$

$$\text{or } m_1 \frac{u \left(1 - \frac{v^2}{c^2} \right)}{\left(1 + \frac{uv}{c^2} \right)} = m_2 \frac{u \left(1 - \frac{v^2}{c^2} \right)}{\left(1 - \frac{uv}{c^2} \right)}$$

$$\text{or } \frac{m_1}{m_2} = \frac{1 + \frac{uv}{c^2}}{1 - \frac{uv}{c^2}} \quad \text{--- (3)}$$

Let us consider the value of the term

$$1 - \frac{u^2}{c^2} = 1 - \frac{\left(\frac{u+v}{1 + \frac{uv}{c^2}} \right)^2}{c^2}$$

$$= 1 - \frac{\left(\frac{u+v}{c} \right)^2}{\left(1 + \frac{uv}{c^2} \right)^2}$$

$$= \frac{1 + \frac{u^2 v^2}{c^4} + \frac{2uv}{c^2} - \frac{u^2}{c^2} - \frac{v^2}{c^2} - \frac{2uv}{c^2}}{\left(1 + \frac{uv}{c^2} \right)^2}$$

$$= \frac{\left(1 - \frac{u^2}{c^2} \right) - \frac{v^2}{c^2} \left(1 - \frac{u^2}{c^2} \right)}{\left(1 + \frac{uv}{c^2} \right)^2}$$

$$= \frac{\left(1 - \frac{u^2}{c^2} \right) \left(1 - \frac{v^2}{c^2} \right)}{\left(1 + \frac{uv}{c^2} \right)^2} \quad \text{--- (4)}$$

$$\text{Similarly } 1 - \frac{u_2^2}{c^2} = \frac{\left(1 - \frac{u^2}{c^2} \right) \left(1 - \frac{v^2}{c^2} \right)}{\left(1 - \frac{uv}{c^2} \right)^2} \quad \text{--- (5)}$$

From eqⁿ (4) and (5), we have

$$\frac{1 - \frac{u_2^2}{c^2}}{1 - \frac{u_1^2}{c^2}} = \frac{\left(1 + \frac{uv}{c^2}\right)^2}{\left(1 - \frac{uv}{c^2}\right)^2}$$

$$\text{or } \frac{\sqrt{1 - \frac{u_2^2}{c^2}}}{\sqrt{1 - \frac{u_1^2}{c^2}}} = \frac{1 + \frac{uv}{c^2}}{1 - \frac{uv}{c^2}} \quad \text{--- (6)}$$

From eqⁿ (3) and (6) we have,

$$\frac{m_1}{m_2} = \frac{\sqrt{1 - \frac{u_2^2}{c^2}}}{\sqrt{1 - \frac{u_1^2}{c^2}}}$$

$$\text{or } m_1 \sqrt{1 - \frac{u_1^2}{c^2}} = m_2 \sqrt{1 - \frac{u_2^2}{c^2}} \quad \text{--- (7)}$$

From equation (7), it is clear that L.H.S. and R.H.S are independent of one another and this result may be true if each is a constant.

$$\text{Therefore, } m_1 \sqrt{1 - \frac{u_1^2}{c^2}} = m_2 \sqrt{1 - \frac{u_2^2}{c^2}} = m_0$$

where m_0 is a constant called the rest mass of the body and corresponds to zero velocity.

$$\text{Thus, } m_1 = \frac{m_0}{\sqrt{1 - \frac{u_1^2}{c^2}}}$$

$$m_2 = \frac{m_0}{\sqrt{1 - \frac{u_2^2}{c^2}}}$$

If m be the mass of the body moving with a velocity u , then

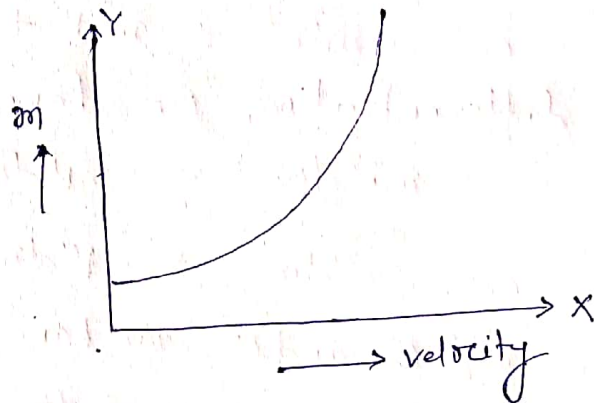
$$\boxed{m = \frac{m_0}{\sqrt{1 - \frac{u^2}{c^2}}}} \quad \text{--- (8)}$$

Equation (8) is the relativistic formula for the variation of mass with velocity. When $u \rightarrow c$, then $m \rightarrow \infty$ i.e. an object travelling at a speed of light would have infinite mass.

Thus, no material particle can have a velocity equal to or greater than the velocity of light.

At ordinary velocity i.e. $v \ll c$, $\frac{v^2}{c^2}$ may be neglected

Thus $m = m_0$.



EINSTEIN'S MASS-ENERGY RELATION

Let us consider a particle of mass m acted upon by a force F in the same direction as its velocity v and the force displaced the body through a distance dx . Then the increase in the kinetic energy dE_k of the body is equal to the work done Fdx .

Force is defined as the rate of change of momentum

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv)$$

According to the theory of relativity, both mass and velocity are variable. Therefore,

$$F = \frac{d}{dt}(mv) = m \frac{dv}{dt} + v \frac{dm}{dt} \quad \text{--- (1)}$$

$$\begin{aligned} \text{Now } dE_k &= Fdx \\ &= m \frac{dv}{dt} dx + v \frac{dm}{dt} dx \\ &= m \frac{dx}{dt} dv + v \frac{dx}{dt} dm \end{aligned}$$

$$\text{or, } dE_k = mv dv + v^2 dm \quad \text{--- (2)}$$

According to the law of variation of mass with velocity

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\text{or } m^2 = \frac{m_0^2}{1 - \frac{v^2}{c^2}}$$

$$\text{or } m^2(c^2 - v^2) = m_0^2 c^2$$

$$\text{or } m^2 c^2 = m^2 v^2 + m_0^2 c^2$$

On differentiation, we have

$$2mc^2 dm = 2mv^2 dm + 2m^2 v dv + 0$$

$$\text{or } c^2 dm = v^2 dm + m v dv \quad \text{--- (3)}$$

From eqⁿ (2) and (3), we have

$$dE_k = c^2 dm \quad \text{--- (4)}$$

Thus, a change in kinetic energy dE_k is directly proportional to a change in mass dm . When a body is at rest its velocity is zero ($K.E = 0$) and mass is m_0 . When its velocity is v , its mass becomes m . Therefore, integrating eqⁿ (4), we have

$$\int_0^{E_k} dE_k = c^2 \int_{m_0}^m dm$$

$$\text{or, } \boxed{E_k = mc^2 - m_0 c^2} \quad \text{--- (5)}$$

This is the relativistic formula for kinetic energy. When the body is at rest, the internal energy stored in the body is $m_0 c^2$. $m_0 c^2$ is called the rest mass energy.

Thus, the total energy E of the body is the sum of kinetic energy and rest mass energy.

$$E = E_k + m_0 c^2$$

$$= mc^2 - m_0 c^2 + m_0 c^2$$

$$\text{or } \boxed{E = mc^2} \quad \text{--- (6)}$$

This is Einstein's mass-energy relation.

This relation states a universal equivalence between mass and energy. It means that mass may appear as energy and energy as mass.

The formula for K.E. reduces to the classical formula for $v \ll c$

$$E_K = mc^2 - m_0c^2$$

$$= \frac{m_0c^2}{(1 - \frac{v^2}{c^2})^{1/2}} - m_0c^2$$

$$= m_0c^2 \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} - 1 \right]$$

$$= m_0c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} - 1 \right)$$

$$= \frac{1}{2} m_0 v^2$$

(From Binomial theorem. For $v \ll c$ higher terms neglected)

RELATIVISTIC RELATION BETWEEN ENERGY AND MOMENTUM

Total energy $E = mc^2$

Momentum $p = mv$

Now $m = \frac{m_0}{(1 - \frac{v^2}{c^2})^{1/2}}$

or $m^2 = \frac{m_0^2}{1 - \frac{v^2}{c^2}}$

or $m^2 = \frac{m_0^2 c^2}{c^2 - v^2}$

or $m^2 c^2 - m^2 v^2 = m_0^2 c^2$

or $m^2 c^4 - m^2 v^2 c^2 = m_0^2 c^4$ (multiplying by c^2)

or $E^2 - p^2 c^2 = m_0^2 c^4$

or $E^2 = p^2 c^2 + m_0^2 c^4$

MASSLESS PARTICLE

Photons are packets of energy which are emitted by a source of radiation and travel with the speed of light. mass of the particle moving with velocity v

is given by

$$m = \frac{m_0}{(1 - \frac{v^2}{c^2})^{1/2}}$$

or $m_0 = m \left(1 - \frac{v^2}{c^2}\right)^{1/2}$

For photon, $v = c$, hence $m_0 = 0$

Thus, the rest mass of photon is zero. It cannot exist at rest.

EXAMPLES OF ENERGY AND MASS EQUIVALENCE

1. $E = mc^2$ explains nuclear reactions such as fission and fusion. These reactions take place in nuclear bombs and reactors. When a uranium nucleus is split up, the decrease in its total rest mass appears in the form of an equivalent amount of K.E. of its fragments.
2. $E = mc^2$ explains binding energy of nucleus.
3. When an electron and a positron combine and disappear, we find high energy radiation called γ -radiation whose radiant energy is exactly equal to the rest mass and kinetic energy of the disappearing particles. This process is reversible and when γ -radiation disappears, we get electron and positron whose total energy i.e. rest mass and kinetic energy is equal to the radiant energy lost.